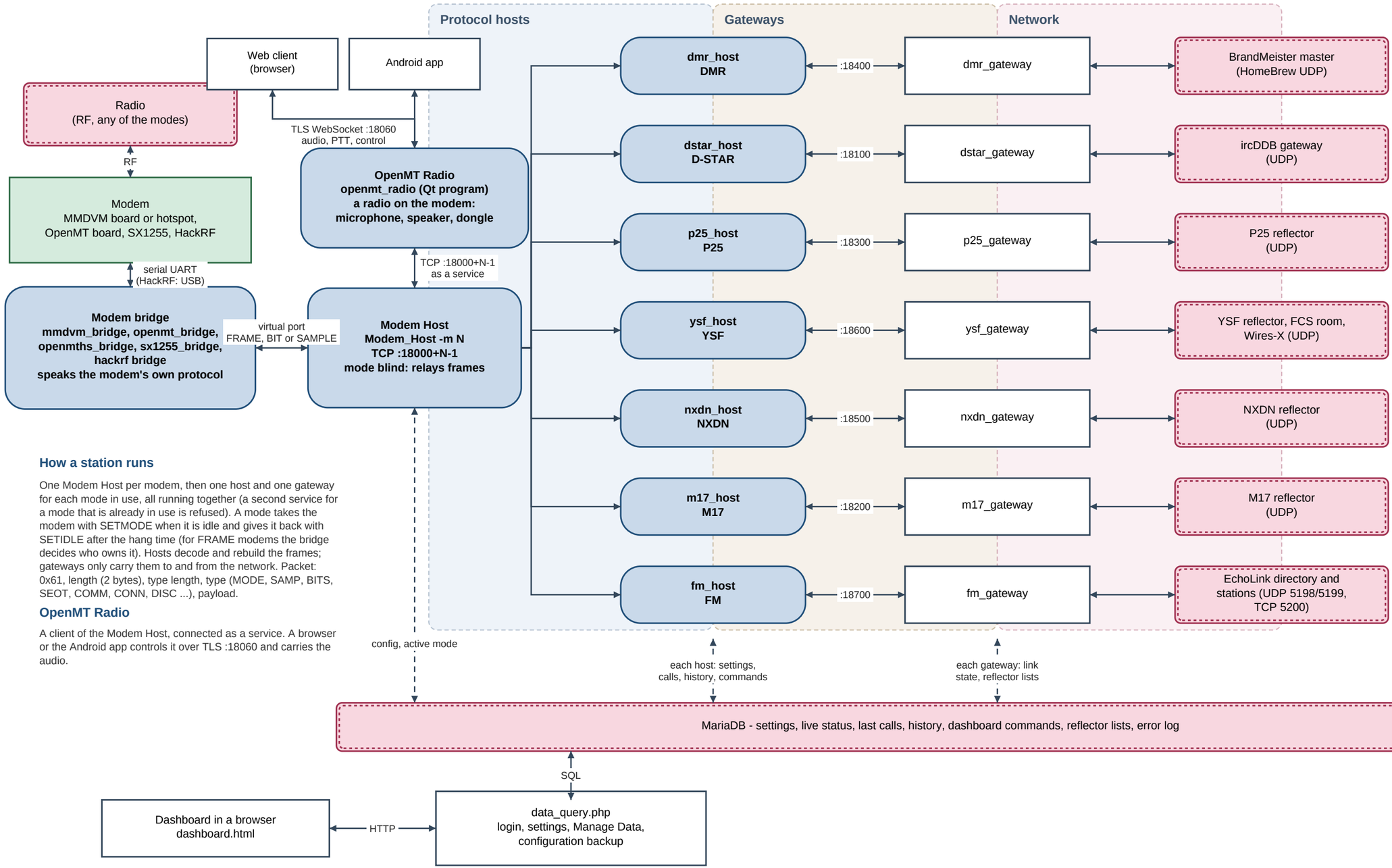
OpenMT - the whole system

Every program that runs on a station, and how data moves between them. N is the modem number (1 to 10); every port below is shown for modem 1.



How a station runs

One Modem Host per modem, then one host and one gateway for each mode in use, all running together (a second service for a mode that is already in use is refused). A mode takes the modem with SETMODE when it is idle and gives it back with SETIDLE after the hang time (for FRAME modems the bridge decides who owns it). Hosts decode and rebuild the frames; gateways only carry them to and from the network. Packet: 0x61, length (2 bytes), type length, type (MODE, SAMP, BITS, SEOT, COMM, CONN, DISC ...), payload.

OpenMT Radio

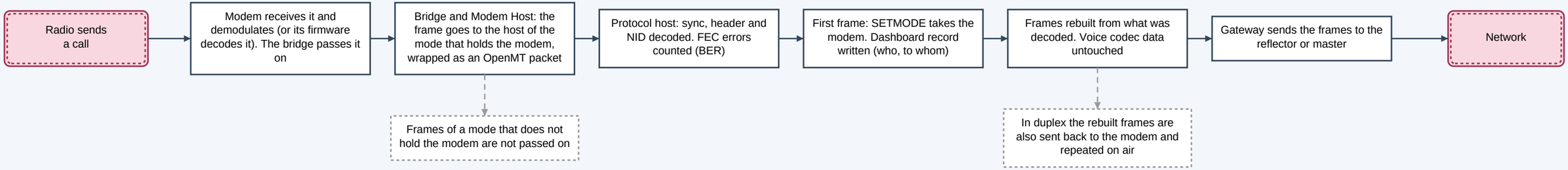
A client of the Modem Host, connected as a service. A browser or the Android app controls it over TLS :18060 and carries the audio.

Reflector links are made from the dashboard: the command goes into the database, the host reads it, tells its gateway to link or unlink, and writes the result back.
With -m N each port is 18000+N-1 (host to Modem Host), 18100+N-1 (D-STAR), 18200+N-1 (M17), 18300+N-1 (P25), 18400+N-1 (DMR), 18500+N-1 (NXDN), 18600+N-1 (YSF), 18700+N-1 (FM). OpenMT Radio: 18000+N-1 to the Modem Host, 18060 for remote control (TLS), UDP 18050 / 18051 to the HackRF / SX1255 bridge.

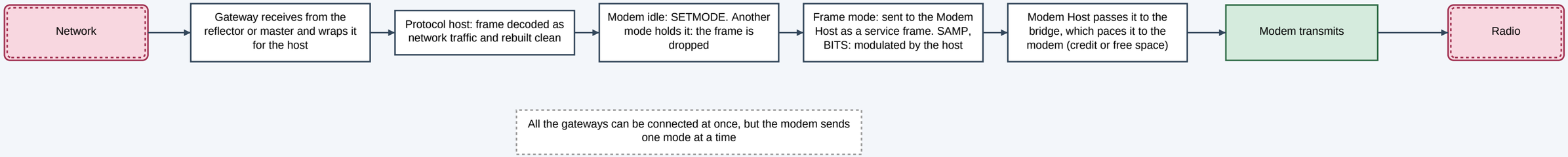
OpenMT - one call through the system

A call from the radio to the network, a call from the network to the radio, and how a call ends and gives the modem back.

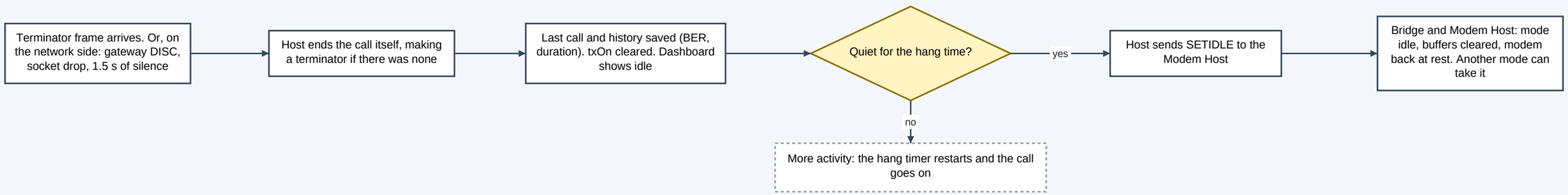
Radio -> network



Network -> radio



End of a call (either direction)



A transmission that runs too long is ended by a timeout (in the bridge for FRAME modems, in the Modem Host for the others), which also sets the mode idle. Errors anywhere go to the error log table with reportError().